ELECTRICAL LOAD SCHEDULE & MAXIMUM DEMAND REPORT

Metropolis Tower (12-Storey Commercial Office)

Commercial high-rise with multi-zone sub-boards, tenant distribution, VRV/VFD HV...

Date: 11/1/2026
Standard: 400V / 230V 50Hz (IEC Standard)
Voltage: 400V / 230V · 50Hz
Engineer: Senior Electrical PE

1. MAIN SWITCH BOARD (MSB): MSB-01 - Main Low Voltage Switchboard

Total Plant MD	Main Incomer	Transformer	Generator & PFC
112.8 kW	**250A MCCB**	**200 kVA**	**200 kVA Prime**
128.4 kVA	Ib = 185 A	64.2% Operating Load	Capacitor: 110 kVAr

MSB Phase	Connected (kW)	Connected (kVA)	Demand (kW)	Demand (kVA)	Current (A)	Neutral / Unbalance
Phase R (Red)	59.9	68.4	36.2	41.2	179.0 A	
Phase Y (Yellow)	60.5	68.4	37.9	43.0	186.9 A	Neutral: 11.5 A
Phase B (Blue)	66.0	75.5	38.7	44.2	192.1 A	Unbalance: 7.1% (OPTIMAL)

PFC Capacitor Bank Sizing & Steps Schedule (110 kVAr @ 525V | Main Incomer: 150A ACB/MCCB)

Step #	Rated Q (@525V)	Eff Q (@420V)	Qty	Total Rated Q	Total Eff Q	Op Current	Step Breaker
1	5 kVAr	3.2 kVAr	4 units	20 kVAr	12.8 kVAr	6.6 A	15A 3P MCCB
2	10 kVAr	6.4 kVAr	3 units	30 kVAr	19.2 kVAr	13.2 A	15A 3P MCCB
3	20 kVAr	12.8 kVAr	3 units	60 kVAr	38.4 kVAr	26.4 A	30A 3P MCCB
TOTAL	—	—	10 active	110 kVAr	70 kVAr	FLC @ 420V	150A 3P ACB/MCCB

2. SUB-DISTRIBUTION BOARDS (SB) FEEDER SCHEDULE

#	Sub-DB Code	Description / Zone	Fed From	CL (kW)	DF	MD (kW)	MD (kVA)	Ib (A)	Incomer Breaker	Feeder Cable & VD	Genset
1	Branch_C-01	Podium & Retail Sub-Distribution Board	msb-main	4.3	0.90	3.4	3.7	15.9 A	75A 3-Pole MCCB	4x1C 35.0mm² Cu/PVC +E (35m, 0.18% · <= 4% OK)	Normal
2	SB-TOWER	Tower Office Floors Sub-Distribution Board	msb-main	103.4	0.90	71.2	78.9	127.3 A	200A 3-Pole MCCB	4x1C 120.0mm² Cu/XLPE/PVC +E (35m, 0.16% · <= 4% OK)	Fire Safety

3. FINAL DISTRIBUTION BOARDS & CIRCUIT SCHEDULES

1. CP-LIFT: Lift Control Panel [3-Phase (4W) · Fire Life-Safety]

MD: 66.37 kW (112.8 A)

Incomer: 150A 3-Pole MCCB · Fed by: Direct MSB (MSB-01) · Board DF: 100% · Margin: +30% · Unbalance: 0.0% (OPTIMAL)
Feeder Cable: 4x1C 95.0mm² Cu/XLPE/PVC +E (25m) · VD: 0.12% (0.47V) · Limit: <= 4.00% [PASSED]

#	Circuit Description	Category	Ph	Qty	W/unit	PF	DF	CL (kW)	MD (kW)	Ib (A)	Breaker	Cable Size
1	Lifts	CUSTOM	3P	3	25000	0.85	0.85	75.00	63.75	108.3 A	125A Type-C 3P MCCB	4x1C 70.0mm² Cu/XLPE/PVC +E
2	Exhaust fan	HVAC	3P	3	100	0.82	1.00	0.30	0.30	0.5 A	16A Type-C 3P MCB	4x1C 2.5mm² Cu/PVC +E
3	Lighting Circuit 1	LIGHTING	Y	20	36	0.95	0.85	0.72	0.61	2.8 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
4	Lighting Circuit 2	LIGHTING	B	20	36	0.95	0.85	0.72	0.61	2.8 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
5	Lighting Circuit 3	LIGHTING	R	20	36	0.95	0.85	0.72	0.61	2.8 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
6	Power Circuit 1	RECEPTACLE	Y	2	200	0.85	0.40	0.40	0.16	0.8 A	6A Type-C 1P MCB	2x1C 1.5mm² Cu/PVC +E
7	Power Circuit 2	RECEPTACLE	B	2	200	0.85	0.40	0.40	0.16	0.8 A	6A Type-C 1P MCB	2x1C 1.5mm² Cu/PVC +E
8	Power Circuit 3	RECEPTACLE	R	2	200	0.85	0.40	0.40	0.16	0.8 A	6A Type-C 1P MCB	2x1C 1.5mm² Cu/PVC +E
	TOTAL FOR CP-LIFT							**78.66**	**66.37**	**112.8 A**	**150A 3-Pole MCCB**	**4x1C 95.0mm² Cu/XLPE/PVC +E**

2. DB-CP-1F: Carpark First Floor [1-Phase (Phase R + N)]

MD: 3.82 kW (17.6 A)

Incomer: 25A 1-Pole MCB · Fed by: Branch_C-01 · Board DF: 100% · Margin: +30% · Phase R @ 230V
Feeder Cable: 2x1C 6.0mm² Cu/PVC +E (25m) · VD: 0.87% (2.00V) · Limit: <= 3.82% [PASSED] · Total: 1.05% / 4%

#	Circuit Description	Category	Ph	Qty	W/unit	PF	DF	CL (kW)	MD (kW)	Ib (A)	Breaker	Cable Size
1	Light Circuit 1 (T8 4'x2)	LIGHTING	1P	15	36	0.95	1.00	0.54	0.54	2.5 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
2	Light Circuit 2 (T8 4'x2)	LIGHTING	1P	15	36	0.95	1.00	0.54	0.54	2.5 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
3	Light Circuit 3 (T8 4'x2)	LIGHTING	1P	15	36	0.95	1.00	0.54	0.54	2.5 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E

#	Circuit Description	Category	Ph	Qty	W/unit	PF	DF	CL (kW)	MD (kW)	Ib (A)	Breaker	Cable Size
4	Light Circuit 4 (T8 4'x2)	LIGHTING	1P	15	36	0.95	1.00	0.54	0.54	2.5 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
5	Light Circuit 5 (T8 4')	LIGHTING	1P	20	18	0.95	1.00	0.36	0.36	1.6 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
6	Light Circuit 6 (T8 4')	LIGHTING	1P	20	18	0.95	1.00	0.36	0.36	1.6 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
7	Light Circuit 7 (Wall Light)	LIGHTING	1P	8	42	0.95	1.00	0.34	0.34	1.5 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
8	Light Circuit 8 (Wall Light)	LIGHTING	1P	8	35	0.95	1.00	0.28	0.28	1.3 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
9	Power Circuit 1 (13A S/S/O)	RECEPTACLE	1P	2	200	0.85	0.40	0.40	0.16	0.8 A	6A Type-C 1P MCB	2x1C 1.5mm² Cu/PVC +E
10	Power Circuit 2 (13A S/S/O)	RECEPTACLE	1P	2	200	0.85	0.40	0.40	0.16	0.8 A	6A Type-C 1P MCB	2x1C 1.5mm² Cu/PVC +E
	TOTAL FOR DB-CP-1F							**4.30**	**3.82**	**17.6 A**	**25A 1-Pole MCB**	**2x1C 6.0mm² Cu/PVC +E**

3. DB-TOWER-L5: Tower Level 5 Corporate Tenant Offices [3-Phase (4W)]

MD: 22.81 kW (46.5 A)

Incomer: 63A 3-Pole MCB · Fed by: SB-TOWER · Board DF: 100% · Margin: +30% · Unbalance: 42.1% (POOR)
Feeder Cable: 4x1C 25.0mm² Cu/PVC +E (25m) · VD: 0.15% (0.61V) · Limit: <= 3.84% [PASSED] · Total: 0.31% / 4%

#	Circuit Description	Category	Ph	Qty	W/unit	PF	DF	CL (kW)	MD (kW)	Ib (A)	Breaker	Cable Size
1	Acoustic Baffle Recessed LED Fixtures	LIGHTING	R	60	38	0.95	0.85	2.28	1.94	8.9 A	10A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
2	Executive Boardroom Preset Dimmers	LIGHTING	Y	24	25	0.95	0.80	0.60	0.48	2.2 A	6A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
3	Dual Desk Sockets Ring 1-30	RECEPTACLE	B	30	350	0.85	0.40	10.50	4.20	21.5 A	25A Type-C 1P MCB	2x1C 6.0mm² Cu/PVC +E
4	Dual Desk Sockets Ring 31-60	RECEPTACLE	R	30	350	0.85	0.40	10.50	4.20	21.5 A	25A Type-C 1P MCB	2x1C 6.0mm² Cu/PVC +E
5	Air Handling Unit AHU-L5 Fan Motor	HVAC	3P	1	11000	0.84	0.85	11.00	9.35	16.1 A	20A Type-C 3P MCB	4x1C 4.0mm² Cu/PVC +E
6	Executive Tea Bar & Water Dispensers	POWER_APPLIANCE	Y	2	2200	0.95	0.60	4.40	2.64	12.1 A	16A Type-C 1P MCB	2x1C 2.5mm² Cu/PVC +E
	TOTAL FOR DB-TOWER-L5							**39.28**	**22.81**	**46.5 A**	**63A 3-Pole MCB**	**4x1C 25.0mm² Cu/PVC +E**

4. DB-TOWER-L10: Tower Level 10 High-Density Trading Floor & Data Hub [3-Phase (4W)]

MD: 56.30 kW (103.8 A)

Incomer: 150A 3-Pole MCCB · Fed by: SB-TOWER · Board DF: 100% · Margin: +30% · Unbalance: 50.3% (POOR)
Feeder Cable: 4x1C 95.0mm² Cu/XLPE/PVC +E (25m) · VD: 0.12% (0.47V) · Limit: <= 3.84% [PASSED] · Total: 0.28% / 4%

#	Circuit Description	Category	Ph	Qty	W/unit	PF	DF	CL (kW)	MD (kW)	Ib (A)	Breaker	Cable Size
1	Trading Floor Low-Glare LED Troffers	LIGHTING	R	40	45	0.95	0.90	1.80	1.62	7.4 A	10A Type-B 1P MCB	2x1C 1.5mm² Cu/PVC +E
2	Quad-Monitor Trading Pod Desks 1-16	RECEPTACLE	Y	16	900	0.90	0.75	14.40	10.80	52.2 A	63A Type-C 1P MCB	2x1C 25.0mm² Cu/PVC +E
3	Quad-Monitor Trading Pod Desks 17-32	RECEPTACLE	B	16	900	0.90	0.75	14.40	10.80	52.2 A	63A Type-C 1P MCB	2x1C 25.0mm² Cu/PVC +E
4	Dedicated Precision Server UPS Input Feeder	IT_CRITICAL	3P	1	25000	0.96	1.00	25.00	25.00	37.6 A	40A Type-C 3P MCB	4x1C 16.0mm² Cu/PVC +E
5	Computer Room Air Handler CRAH-1	HVAC	3P	1	8500	0.84	0.95	8.50	8.07	13.9 A	16A Type-C 3P MCB	4x1C 2.5mm² Cu/PVC +E
	TOTAL FOR DB-TOWER-L10							**64.10**	**56.30**	**103.8 A**	**150A 3-Pole MCCB**	**4x1C 95.0mm² Cu/XLPE/PVC +E**